Jacomo Corrieri

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EDUCATION

University of Florida, *Gainesville, FL* *Expected May 2027*
Master of Science (M.S.) in Computer Science
GPA: 3.83/4.0

University of North Florida, *Jacksonville, FL* *May 2025*
Bachelor of Science (B.S.) in Computer Science, Minors in Mathematics and International Business
GPA: 3.99/4.0 | *Summa Cum Laude* | Hicks Honors College

WORK EXPERIENCE

Florida Blue *May 2026 – Present*
Information Technology Intern – Software Engineering
• Develop an application to manage, standardize, and track relational migration resources using React.js/Express.js and Java Spring Boot, targeting **8–10 hours** of weekly time savings by eliminating manual database operations.
• Validate backend correctness with SQLite-based unit tests, benchmark SQL Server query response times, and test deployments using Jenkins and OpenShift while maintaining system design documentation and Agile user stories.
• Implement SQL Server transactions with pessimistic locking and timestamp-based optimistic concurrency checks to prevent conflicting updates across multi-record resources.

UFIT Research Computing *Jan 2026 – Aug 2026*
Research Computing Support Intern
• Built a Python CLI and Django REST API to query **800M+ HiPerGator usage records**; containerized the API and PostgreSQL database with Podman Quadlet and secured endpoints using RS256 JWT authentication.
• Accelerated HPC module deprecation checks by **10–30×** by caching read-heavy filesystem module data in PostgreSQL tables, refreshed hourly via Slurm cron jobs, and automated daily usage data ingestion for **3700+ users**.

RESEARCH

ICTAI 2026 - *Scalable Multi-robot Coverage with Connectivity Maintenance Via Deep Reinforcement Learning*
Corrieri Jacomo, Kulbaka Iliya, Ghosh Anirban, Dutta Ayan
• Simulated up to 50 autonomous robots with Python and Gymnasium, achieving **5x more scalability** than prior work.
• Developed a decentralized CNN model with PyTorch, resulting in **15% greater area coverage** compared to our baseline.

PROJECTS

Tiny DNS Cache *Sep 2026 – Present*
• Build a multithreaded C++20 DNS sim using Unix sockets, thread pooling, and a bounded producer-consumer queue.
• Design a thread-safe generic cache with concepts and a shared mutex, plus a RAII SQLite wrapper and unit tests.
• Develop record lookup with recursive CTEs for CNAME/NS chains, zero-copy views, and binary IPv4/IPv6 decoding.

MoTorch Neural Network Library *Apr 2026 – Jul 2026*
• Created a PyTorch-inspired deep learning framework with neural network modules, optimizers, and loss functions.
• Implemented a reverse-mode autodiff engine with a dynamic computational graph for backpropagation.
• Automated CI and testing with GitHub Actions, benchmarking across linear models with up to **32.6 million parameters**.

UF Student Marketplace *Feb 2026 – Jul 2026*
• Engineered real-time messaging in Go using WebSockets and concurrent client read/write goroutines.
• Replaced database blobs with scalable cloud-based object storage by integrating AWS S3 with a Gin REST API.
• Designed and indexed relational schemas with GORM and SQLite for listings, messaging, and user workflows.

SKILLS

Languages: Python, C++, Go, Java, TypeScript, SQL
Frameworks & Libraries: FastAPI, Django, Gin, React, PyTorch, NumPy, Scikit-Learn, Gymnasium
Infra & Tools: Linux, Slurm, Git, Podman PostgreSQL, SQLite, GitHub Actions, AWS S3, Claude, Codex, Gemin, CMake